KG VOYAGER TOOLS

SEARCH

Class Search

`search_all_classes(search_term, limit)`

Search for classes in the Knowledge Graph by name or concept. Uses hybrid retrieval (semantic + keyword) to find relevant classes. Low-relevance results are automatically pruned.

Purpose: Use natural-language terms from the question to find candidate class URIs.

INPUT SCHEMA

`search_term` string REQ

The concept, keyword, or class name to search for. Utilizes semantic search to find relevant matches.

`limit` number OPT

Maximum number of results to return (default: 10).

OUTPUT FORMAT

TABLE WITH COLUMNS:

`class_uri` `label` `count` `score`

Instance Lookup

`lookup_class_instances(class_uri, search_term, limit, mode)`

Retrieve instances of a specific class, optionally filtered by name.

Purpose: The primary way to resolve named entities mentioned in the question (e.g., 'France', 'Albert Einstein') to their URIs. First identify the entity's class via class search, then use this tool with the entity name as search_term.

INPUT SCHEMA

`class_uri` string REQ

The full absolute URI of the class. Do not use prefixed/abbreviated URIs.

`search_term` string OPT

Optional keyword to filter entities by label/name.

`limit` number OPT

Maximum number of entities to return (default: 10).

`mode` string OPT

Result mode: 'labels_only' (default) searches only entity label; 'all_attributes' searches across all literal values.

OUTPUT FORMAT

TABLE WITH COLUMNS:

`entity` `displayLabel` `searchProp` `searchVal`

Property Search

`search_all_properties(search_term, limit)`

Search for properties in the Knowledge Graph by name or concept. Uses hybrid retrieval (semantic + keyword) to find relevant properties. Low-relevance results are automatically pruned.

Purpose: Find candidate property URIs. Use the domain and range columns to understand how the property connects different entity types.

INPUT SCHEMA

`search_term` string OPT

The concept, keyword, or property name to search for. Utilizes semantic search to find relevant matches.

`limit` number OPT

Maximum number of results to return (default: 10).

OUTPUT FORMAT

TABLE WITH COLUMNS:

`property_uri` `label` `aliases` `domain` `range` `count` `score`